MCO-1D

Operator Manual — Rev 24

MCO-1D ornithopter SAR drone — September 2026

Document	MCO-1D_Operator_Manual_v24
Revision	September 2026
New in this revision	Flight monitor · AI observer · Soaring variometer · Stability system · Density altitude · 97 learning hypotheses
Audience	Pilot / Operator — no engineering background required
Companion docs	Commissioning Manual v20 · Software Reference v24 · Field Cards v24

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18 Flight Monitor — Health Watchdog

The flight monitor watches the aircraft continuously and speaks up when something needs attention. It has three tiers — immediate alerts you must act on, advisories you should know about, and a silent log available on request. When everything is normal, you hear nothing.

◆ **NOTE** The monitor cannot control the aircraft. It can only tell you what it sees. You make the decisions.

What it watches

Battery	Warns at 21.5V. Critical at 20V (begin return). Land now at 19V.
Power	Warns if sustained above 3.5 kW. Critical above 4.5 kW.
Vibration	Warns at 0.15g RMS. Structural concern at 0.35g — land and inspect.
Wing guard	Warns if firing more than 3 times/minute. Critical above 10/min.
Trim drift	Warns if VSI drifts more than 50 ft/min at neutral controls.
Roll asymmetry	Warns if persistent roll above 2.5° at neutral controls.
Rope fatigue	Warns if rope life burning faster than 2%/minute.
Oscillation	Immediate alert if VSI reverses more than 3 times in 2 seconds.
Power vs speed	Warns if power is 40% above expected for current airspeed.
Battery drain	Warns if voltage dropping faster than 2V/minute.

Silence rules

The monitor waits at least 15 seconds between any two alerts. It does not repeat the same alert within 60 seconds. During aerobatic manoeuvres, only immediate (Tier 1) alerts fire.

Monitor commands

"Monitor report / flight health / what has the monitor seen"
→ Full report — alerts, trend data, trim baseline, log.

"Monitor thresholds / what does it watch for"
→ All threshold values for each condition.

"Disable the monitor"
→ Silences all alerts. Not recommended.

"Enable the monitor"
→ Re-enables after disable.

19 AI Observer — Learning Watcher

The AI observer watches the learning system — discovery, the optimizer, and the genome. It looks for patterns in what is working and what is not, forms specific suggestions, and proposes them to you. It speaks up roughly once or twice per flight when it has something worth saying. When it has no suggestion, you hear nothing.

◆ **NOTE** The observer never acts without your permission. It proposes. You decide. If you say yes, it runs a brief test and reports the result. If the test improves the aircraft, it commits the improvement to the learning archive.

What it watches

Trim drift	VSI or roll error at neutral controls → proposes AoA or cascade adjustment.
Discovery plateau	Ps score flat for 12+ minutes → proposes narrowing or broadening search.
Archive coverage	Too many empty niches → proposes broader exploration.
Genome convergence	One mechanism dominating → reports and optionally reinforces.
Cascade clustering	Winning configs clustered in narrow range → proposes refining search.

A typical exchange

◆ **NOTE** Observer: "Trim VSI has been +45 ft/min for the last 3 minutes at neutral controls. A 0.5° AoA reduction should bring it to zero. Want me to try it?" — Say yes or no.

Observer commands

"Yes / go ahead / try it / approve"
→ Runs the suggested test. Reports result within 5-10 seconds.

"No / not now / skip it"
→ Declines the suggestion. Observer will not repeat it.

"Observer report / what has the AI noticed / learning quality"
→ Full report: discovery trend, archive coverage, genome weights, recent proposals.

"How is discovery going / Ps trend / is learning improving"
→ Discovery quality summary.

"Genome status / mechanism weights"
→ All 6 genome axes with weights and trend.

"Disable the observer / stop suggestions"
→ Observer goes silent. Still watches, does not propose.

20 Soaring and Motor Glider Flight

Pull the power lever below 15% and the aircraft becomes a glider. The VSI automatically switches to variometer mode and the soaring system activates. The aircraft behaves like a motor glider with the engine at idle — it glides efficiently, can circle in thermals, and the instruments tell you what the air mass is doing.

The three variometer modes

Basic VSI	Stroke-smoothed altitude rate. Normal powered flight.
Total energy (TE)	Removes speed/altitude trades. Shows true energy state. The pull-up does not fool it — only real lift does.
Netto	Shows only air mass movement. Zero in still air at any speed. Positive = rising air (thermal). The soaring pilot's

◆ **NOTE** Netto auto-activates when power is below 15%. It shows the air mass, not the aircraft. If it reads +0.3 m/s while you are descending, the air is rising at 0.3 m/s — seek more of it.

Glide performance

Best glide speed	~9.4 m/s (18 kt) — fly this to maximise distance to next thermal.
Min sink speed	~7.2 m/s (14 kt) — fly this to maximise time in a thermal.
Max L/D	~15:1 — estimated pre-flight. Refined by discovery in flight.
Min sink rate	~0.53 m/s (104 ft/min) — estimated. Refined in flight.

◆ **NOTE** These values are initial estimates. The discovery system measures the actual polar in flight and refines them over time. After several soaring sorties the MacCready advisory becomes progressively more accurate.

MacCready speed-to-fly advisory

The ground station shows the optimal inter-thermal speed based on the thermal strength the variometer has measured. Stronger thermals → fly faster between them. No expected thermals → fly at best glide speed.

Thermalling

When the netto shows positive for more than 5 seconds, the observer announces the thermal and its strength. Circle at minimum sink speed (~7 m/s) with 40-45° of bank. The aircraft's low wing loading lets it circle tightly in the thermal core where the lift is strongest.

Soaring commands

"Soaring status / variometer reading / is there a thermal"
→ Full soaring status — netto, thermal detected, MacCready advisory.

"Total energy vario / TE vario"
→ Switch to total energy variometer mode.

"Netto variometer / soaring mode / glider mode"
→ Switch to netto variometer. Soaring mode arms.

"Basic VSI / normal VSI"

→ Return to stroke-smoothed basic VSI.

"What speed should I fly / MacCready speed"

→ AI reads MacCready advisory and answers in knots.

"Best glide speed"

→ AI reports best glide speed for current conditions.

21 Stability and Coordinated Turns

The aircraft uses three mechanisms together to track stably and turn without rudder input — the same way a well-designed conventional aircraft does it automatically.

Proverse yaw	Roll input also yaws the nose into the turn at entry. Prevents adverse yaw. The turn starts coordinated.
Weathercock	Cascade phase creates yaw-into-wind tendency. The aircraft aligns with the relative wind automatically. Kee
Dihedral	Wing geometry creates a roll-away-from-sideslip moment. If a gust pushes the wings, the aircraft self-levels

◆ **NOTE** These three mechanisms are optimised together by the learning system as a coordinated unit — the stability triangle. After enough soaring sessions the aircraft will track straight hands-off and turn without the nose swinging out.

What the pilot feels after learning converges

- Bank into a turn — the nose follows into the arc without swinging outward
- Hold the bank — the aircraft tracks around without tightening or flattening
- Release the controls in a bank — the aircraft slowly returns toward wings-level
- Cruise hands-off — heading holds within a few degrees
- Turbulence rocks the wings — they return without pilot input
- High-speed roll — crisper and more responsive than at cruise

22 High Density Altitude Operations

Hot, humid, and high-altitude conditions reduce air density. Less dense air means less lift. The MCO-1D compensates automatically using sensor-fusion density estimation and amplitude scaling — the same strategy bumblebees use to fly at high altitude.

The MCO-1D's advantage

◆ **NOTE** Helicopters lose power AND lift in thin air — a double penalty. The MCO-1D's electric motors produce rated shaft power regardless of air density. Only the aerodynamics change. Full motor authority is available for compensation.

Kabul, summer	1791m / 35°C — 21% less lift than sea level. Amplitude scales from 96° to ~106°. Within limits.
Mountain pass	3000m / 30°C — 28% less lift. Amplitude ~113°. Within limits.
Extreme high pass	4500m / 25°C — 38% less lift. Amplitude 115° + slight freq increase.

What the system does

The density estimator runs continuously from pitot and IMU. When it detects thin air, it scales amplitude first (the bumblebee strategy). Frequency only increases when amplitude alone is insufficient. Cascade phase is re-optimised at each density band separately — the LEV capture timing changes in thin air.

Density altitude commands

"Density altitude status / thin air / altitude compensation"

→ Current density estimate, compensation band, amplitude scale factor.

Appendix A Useful Phrases — Complete

Starting

"Ready to fly"
→ Pre-flight checks.

"Arm it"
→ Arms after confirmation.

"We're done / land and save"
→ Saves learning, disarms.

Reference bugs

"Target 100 ft/min down / descent rate 100"
→ vsi_ref -100.

"Set heading bug to 090"
→ hdg_ref 090.

"Altitude bug 1000 feet"
→ alt_ref 1000.

"Clear the VSI / heading / altitude bug"
→ vsi_ref_clear / hdg_ref_clear / alt_ref_clear.

"Precision status"
→ All three bugs plus trim quality.

Flight monitor

"Monitor report / flight health"
→ Full monitor report.

"Monitor thresholds"
→ What triggers each alert tier.

"Disable / enable the monitor"
→ Silence or restore alerts.

AI observer

"Yes / go ahead / approve"
→ Run the suggested test.

"No / not now / skip"
→ Decline the suggestion.

"Observer report / learning quality / what has the AI noticed"
→ Full observer report.

"How is discovery going / Ps trend"
→ Discovery quality summary.

"Genome status / mechanism weights"
→ All 6 genome axes.

"Disable the observer"
→ Observer goes silent.

Soaring

"Soaring status / is there a thermal"
→ Full soaring status.

"Netto variometer / soaring mode"
→ Netto vario active.

"Total energy vario"
→ TE variometer active.

"Basic VSI"
→ Return to normal VSI.

"MacCready speed / what speed should I fly"
→ Speed advisory.

"Best glide speed"
→ Best glide speed for conditions.

Stability

"Precision flying / fly steady"
→ Precision focus — trims stability.

Density altitude

"Density altitude status / thin air"
→ DA compensation status.

Discovery

"Discovery registry / show all hypotheses"
→ All 97 hypotheses + status.

"Discovery focus list"
→ All mission foci.

"Mechanism weights"
→ Genome report.

Speed and performance

"Top speed / as fast as possible"
→ Top speed focus.

"Fastest climb / clear the trees"
→ Climb rate / angle focus.

"Aerobatics / tight turns"
→ Aerobatics focus.

"Back to normal"
→ Clears focus.

Wing coordination

"Enable clap and fling"
→ ~25% hover lift enhancement.

"Anti-phase cruise"
→ 115° amplitude mode.

ALSS

"Start practice flights"
→ ALSS starts.

"Stop the practice"
→ Ends after current sortie.

"Abort"
→ Immediate landing.

Ceiling

"Arm ceiling exploration"
→ Step 1.

"Confirm ceiling exploration"
→ Step 2 — begins probe.

"Abort ceiling test"
→ Descends immediately.

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Revision 24 — September 2026

New in v24: Flight monitor (§18), AI observer (§19), Soaring / motor glider flight (§20), Stability and coordination (§21), High density altitude (§22). All 97 hypotheses integrated into discovery, optimizer, and genome. Six genome axes. Thirty MAP-Elites intent bins.